

Project Design Phase

Proposed Solution Template

Date	30 September 2026
Team ID	SWTID-2026-5226
Project Name	Study Buddy API
Maximum Marks	2 Marks

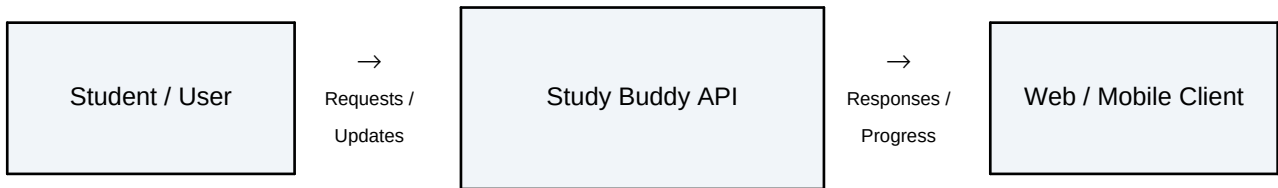
Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Students often use separate tools for planning study schedules, managing subjects, tracking tasks, and accessing study resources. This fragmentation can make it difficult to organize learning activities and monitor progress. Study Buddy API provides a common backend service for student-focused study planning and productivity features.
2	Idea / Solution description	Study Buddy API is a REST-based backend that manages users, subjects, study plans, tasks, study sessions, progress, and study resources. It exposes authenticated API endpoints that a web or mobile application can use to create study plans, update tasks, record sessions, retrieve progress, and receive personalized study-related data.
3	Novelty / Uniqueness	The solution brings core study-management functions into a single API layer with modular resources and reusable endpoints. The design can support personalized study plans, progress tracking, and integration with different front-end clients without coupling the backend to one user interface.
4	Social Impact / Customer Satisfaction	A centralized study-management service can help students organize academic work, maintain consistent study routines, and understand their progress. Clear APIs and structured data can also make the system easier to extend with accessibility features and additional learning tools.
5	Business Model (Revenue Model)	A possible model is freemium access: basic study planning and tracking can be available at no cost, while advanced analytics, larger resource storage, premium integrations, or institution-level features can be offered as paid services. The API can also be licensed or deployed for educational institutions.
6	Scalability of the Solution	The API can be scaled horizontally by running multiple backend instances behind a load balancer. A modular service design, indexed database, caching, pagination, background jobs, and containerized deployment can support increasing users and API traffic. New resources and endpoints can be added without redesigning the complete system.

Data Flow Diagrams – Study Buddy API

DFD Level 0 – Context Diagram



External entities interact with the Study Buddy API through authenticated HTTP requests. The API validates requests, applies business rules, stores or retrieves data, and returns structured responses to the client.

DFD Level 1 – Main Data Flows



Primary data stores: User accounts, subjects/tasks, study plans, study sessions/progress, and study resources. Authentication data should be protected and sensitive credentials should not be stored in plain text.

User Stories – Study Buddy API

The following stories describe expected interactions with the API from the perspective of a student or system administrator.

ID	User Story	Acceptance Criteria
US-01	As a student, I want to register an account so that I can securely use Study Buddy.	Valid registration data creates an account; duplicate/invalid data is rejected; credentials are handled securely.
US-02	As a student, I want to log in so that I can access my study data.	Valid credentials return an authenticated session/token; invalid credentials return an appropriate error.
US-03	As a student, I want to create and manage subjects so that my study plan is organized by subject.	Subject can be created, viewed, updated, and deleted only by an authorized user.
US-04	As a student, I want to create study tasks with deadlines so that I can track what I need to complete.	Task supports subject, title, priority/status, and due date; updates are persisted and retrievable.
US-05	As a student, I want to create a study plan so that I can schedule learning activities.	Plan can be created and retrieved; planned activities are associated with the correct user and subject.
US-06	As a student, I want to record a study session so that my study time is tracked.	Session stores relevant date/time, duration, and subject/activity information and contributes to progress data.
US-07	As a student, I want to view my progress so that I can understand my study activity.	API returns summarized progress based on stored tasks and study sessions for the authenticated user.
US-08	As a student, I want to manage study resources so that useful materials are easy to find.	Authorized users can add, view, update, or remove resource metadata according to permissions.
US-09	As an administrator, I want controlled access to system data so that users and services remain secure.	Protected endpoints require authentication/authorization and unauthorized requests are rejected.

Study Buddy API – Functional Scope

Module	Representative API capabilities
Authentication	Register, login, token/session handling, authorization.
Users	Retrieve and update user profile information.
Subjects	Create, list, update, and delete subjects.
Tasks	Create tasks, set due dates/priorities, update status, list pending/completed tasks.
Study Plans	Create schedules/plans and retrieve planned study activities.
Study Sessions	Start/record sessions, store duration/activity, retrieve history.
Progress	Calculate or retrieve study-time, task-completion, and activity summaries.
Resources	Store and retrieve study-resource metadata and links.

Non-functional considerations

- Security: authenticated endpoints, authorization checks, input validation, secure credential storage, and protected sensitive data.
- Reliability: consistent HTTP status codes, validation errors, transaction handling, and appropriate logging.
- Performance: pagination for large collections, indexed queries, caching where useful, and efficient database access.
- Maintainability: modular controllers/services, versioned APIs, documented endpoints, and automated tests.